

Research paper

Additive manufacturing of intricate lattice materials: Ensuring robust strut additive continuity to realize the design potential

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ABSTRACT

Additive manufacturing (AM) has enabled the emergence of intricate lattice materials with controlled structural or functional properties unfeasible for conventional materials. A striking feature of these novel materials is the substantial number of constituent strut elements contained in them, e.g., more than 5×10^5 struts (diameter: 0.5 mm; length: 5.0 mm) in a common tetrahedron lattice with dimensions of $200 \times 200 \times 200 \text{ mm}^3$. A prerequisite for realizing the full design potential of these structures is to ensure the robust additive continuity of each constituent strut, which refers to geometric continuity between two successive layers of a strut, in order to achieve high lattice structural integrity. This critical issue has, however, remained unanswered. Herein we formulate a design-and-manufacture interlocked strut additive continuity model for powder bed fusion (PBF) and apply it to the entire design gamut of Ti-6Al-4 V struts. On this basis, a strut additive continuity threshold value ($\varphi = 0.9$) is proposed and validated with 23,100 inclined struts in Ti-6Al-4 V lattices manufactured by electron beam PBF (EB-PBF). Design maps are then constructed for EB-PBF of lattice materials. Furthermore, a minimum powder-bed density ($0.6\rho_{\text{TD}}$) for powder selection and a powder-specific initial layer thickness ($0.6D_v(90)$) are recommended. This work establishes a necessary basis for the design and manufacture of robust lattice materials by EB-PBF including graded lattices.

1. Introduction

Intricate lattice materials enabled by powder bed fusion (PBF) additive manufacturing (AM) are topologically ordered, three-dimensional open-cell structures [1–3]. They display controlled structural or functional properties that are unattainable for conventional dense or cellular materials [4–8]. Consequently, they have quickly emerged as a class of novel materials for diverse applications, most notably in the form of lattice structures, which comprise interconnected fine struts that are organised in one or more repeating unit cells, with or without graded designs. A remarkable feature of these novel materials is the substantial number of the fine constituent strut elements contained in them. For example, a common tetrahedron lattice of $200 \times 200 \times 200 \text{ mm}^3$ in dimensions can contain nearly 260,000 inclined struts and a similar number of non-inclined struts (strut diameter: 0.5 mm; strut length: 5.0 mm). The structure derives its anticipated properties from its unique architectural design (e.g., the size, geometry, inclination and spatial arrangement of its unit cells and struts), as well as the constituent

material. Hence, to realize the full design potential of these lattice materials, it is essential to ensure the robust additive continuity of each constituent strut in them, which may have different diameters, inclines or cross-section profiles. This also corresponds to significant cost and lead-time savings in industrial-scale production.

To date, a wide variety of metal lattice structures have been designed, additively manufactured and evaluated [4–8], including arterial stents that comprise 100–150 μm fine inclined Nitinol struts, which are as flexible and corrosion-resistant as conventionally manufactured [9,10]. It should be emphasized that the as-manufactured lattice struts often contain a variety of defects or imperfections, which adversely affects lattice properties. These include

- (i) stair-case roughness represented by $\frac{h}{4}\cos\theta$ (h : actual layer thickness; θ : orientation angle) [11], where each step has a width of $\frac{h}{\tan\theta}$, which further serves as a natural base for particle adherence increasing surface roughness;

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- (ii) surface roughness due to uneven melt pool flow and solidification shrinkage, which may be referred to as primary roughness [12];
- (iii) surface roughness due to adhered or sintered particles (excluding upward-facing horizontal surfaces), which may be described as secondary roughness [12];
- (iv) internal defects (pores, voids, lack-of-fusion); and
- (v) noticeable shape distortion or variations in diameter including narrow sections.

In fact, owing to the lack of proper consideration of lattice strut defects and geometric errors, Finite Element (FE) based predictions of metal lattice properties often noticeably deviate from experimental measurements [13–17]. Consequently, efforts have been made to improve the FE modelling of lattice properties by incorporating the influence of strut geometric imperfections. For example, Li et al. [15], Cao et al. [16] and Lozanovski et al. [17] incorporated geometrical imperfections in lattice struts in their FE models and obtained better consistency between simulations and experiments. In addition, Echeta et al. [18] considered surface roughness in the FE modelling and predicted the elastic moduli of lattices, which are comparable to reported experimental data. These efforts improved our understanding of the mechanical properties of AM lattice materials.

Overall, after years of concerted research, metal AM by either laser PBF (L-PBF) or electron beam PBF (EB-PBF) is able to consistently produce highly dense sections with consistent surface finishes. Furthermore, hot isostatic pressing (HIP) can effectively heal the internal defects in lattice struts [19–21]. Lattice strut internal defects are thus manageable and no longer a prime concern. However, because of the small cross section of each thin strut (~ 1 mm in diameter), any severe surface defects or geometric imperfections such as narrow sections can have a marked potential influence on strut or lattice properties. This underlines the importance of manufacturing consistent, uniform lattice struts at each necessary orientation by design to ensure high structural integrity of the lattice structure. In general, the internal defects and surface roughness problems arising from PBF AM, including those in the lattice struts, have been well studied in the last 15 years. The focus of this study is to predict and minimize or prevent the presence of severe geometric defects in lattice struts fabricated by EB-PBF.

In principle, research has identified that the PBF additive manufacturability of a lattice strut is affected by the base material [22–24], strut diameter [25–29], strut length [25,26], inclination angle [25–27], powder characteristics [23,24], and PBF parameters (beam power, layer thickness, scan strategy, scan speed, hatch spacing, beam size) [23,24,26,27,30]. It is a design-and-manufacture interwoven issue. For example, the minimum manufacturable Ti-6Al-4 V strut diameter was found to be 0.3 mm by L-PBF using a SLM Solutions 250HL system at the layer thickness setting (h_0) of 30 μm if inclined at $\theta = 20^\circ$ [31] or 0.267 mm at $\theta = 45^\circ$ [26] while being 0.7 mm by EB-PBF at $h_0 = 100 \mu\text{m}$ and $\theta = 35^\circ$ [32]. A review of recent literature on metal lattices has confirmed that researchers still rely on experimental exploration to determine the manufacturable strut diameters and inclination angles [33–38]. In other words, a generic manufacturability-informed design criterion that can ensure robust additive continuity of each inclined lattice strut remains absent since the first research paper on PBF metal lattices was published in 2005 [39]. This absence has entailed both design uncertainties and manufacture failures for metal lattices, restricting their further development.

This research aims to tackle this issue, i.e., the absence of a generic manufacturability-informed design criterion for PBF of lattice struts. In the following, we first formulate a comprehensive additive continuity model for PBF AM of lattice struts, which quantifies the influence of both the explicit and latent design and manufacture factors on lattice strut additive continuity. This is followed by the establishment of a minimum threshold parameter ($\varphi_{\text{min-threshold}}$) for strut additive continuity. The criterion is then used to inform or proof-check intricate lattice designs for systematic validation. On this basis, all-inclusive additive continuity

maps are constructed for PBF AM of metal lattices. Their rich implications are discussed for both uniform and graded lattice materials, including the identification of a minimum powder-bed density (ρ_{PBD}) for powder selection and a baseline powder-specific layer thickness setting equation ($h_{0\text{-specific}}$).

2. Experiments

Plasma atomized Ti-6Al-4 V powder with $Dv(10) = 52 \mu\text{m}$, $Dv(50) = 71 \mu\text{m}$ and $Dv(90) = 102 \mu\text{m}$ was used, where $Dv(k)$ is the powder size below which $k\%$ of the powder by volume is included. The powder was produced from wire feedstock (supplied from GE AP&C) and has an apparent density of 2.56 g cm^{-3} and a tap density of $\rho_{\text{TAP}} = 2.88 \text{ g cm}^{-3}$. An Arcam A2 system was employed for EB-PBF AM of all samples using default EB-PBF Arcam parameters, specified in Table 1, with an initial layer thickness setting of $h_0 = 50 \mu\text{m}$.

A total of 165 strut samples were additively manufactured in 11 groups, where each group consists of 15 struts in different diameters but with the same inclination angle. The 15 strut design diameters (mm) are 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.5, 2.0, 2.5 and 3.0 with the same strut length of 20 mm, typical of a lattice unit cell size. The 11 inclination angles ($^\circ$) include 90, 75, 60, 45, 35, 25, 20, 15, 10, 5, 0. Supplementary Fig. S1 shows a view of these 165 manufactured strut samples. In this study, the manufactured diameter refers to the thinnest strut section observed, which was measured from scanning electron microscopy (SEM) images (FEI Nova Nano SEM), as highlighted by the red arrows in Fig. S2.

Tetrahedron and truncated octahedron lattice structures with different dimensions ($58 \times 58 \times 87 \text{ mm}^3$; $27 \times 27 \times 40 \text{ mm}^3$; $48 \times 48 \times 48 \text{ mm}^3$) containing inclined struts were designed with and without following the additive continuity design criterion to be established in this study. They were manufactured using the same EB-PBF parameters (Table 1) to assess the proposed additive continuity design criterion. The dimensions of each EB-PBF-fabricated lattice sample were measured with a caliper.

The strut morphology in an as-manufactured tetrahedron lattice sample (dimensions: $27 \times 27 \times 40 \text{ mm}^3$) was analysed using a micro computed tomography ($\mu\text{-CT}$) scanner (GE Phoenix v-tome-x s240), which has a micro-focus X-ray tube and a nano-focus X-ray tube. The micro-focus X-ray tube was used for the analysis, operated at an accelerating voltage of 120 kV and a current of 40 μA . The spatial resolution of the $\mu\text{-CT}$ scanning depends on several factors including X-ray focal spot size, voxel size, number of projection views per rotation of the X-ray tube, and reconstruction algorithms [40–42]. The voxel size depends on the distance of the sample surface to the micro-focus X-ray tube. When other factors are fixed, the closer the sample surface is to the micro-focus X-ray tube, the higher the spatial resolution. The entire lattice sample was analysed by $\mu\text{-CT}$ at a spatial scanning resolution or voxel size of $16.36 \times 16.36 \times 16.36 \mu\text{m}^3$. The minimum reliable defect size is taken as four times the voxel size in each dimension [43–45]. In the $\mu\text{-CT}$ analysis, it is necessary to define an object's surface in its $\mu\text{-CT}$ volume in

Table 1

Parameters for EB-PBF of Ti-6Al-4 V using an Arcam A2 system.

Basic parameters		Strut diameter > 1 mm	Strut diameter ≤ 1 mm
Layer thickness setting		50 μm	
Build plate and temperature		Stainless steel, preheated to 1003 K (730 $^\circ\text{C}$)	
Acceleration voltage		60 kV	
Electron beam spot size		100 μm	
Preheating	Beam current	30 mA	
	Scanning speed	(1–1.3) $\times 10^4$ mm/s	
Melting	Beam current	20 mA	
	Scanning speed	4500 mm/s	
	Line offset	0.1 mm	0.2 mm
	Hatching strategy	Snake	Point

order to determine its shape. Due to discretization errors and optical factors such as X-ray focal spot size and scattering radiation, the μ -CT value blurs around an object's border, making the exact boundary position uncertain [43,45]. Thresholding the μ -CT value for blurring borders can produce a variety of shapes, depending on threshold values [43, 45]. A voxel is the smallest 3D element of the volume (a single voxel can be a noise signal in the μ -CT data). Reducing its size to one-fourth of the feature or defect size reduces the blurring effect and thus increases the shape accuracy determined by the μ -CT data [43,45]. This limits the minimum detectable defect size to $\sim 60 \mu\text{m}$ in this study. The μ -CT images were reconstructed using Phoenix datosx 2 and analysed using VGStudio MAX 3.0 to determine the geometric continuity of each inclined strut.

3. Results

3.1. Formulation of a lattice strut additive continuity model

A metal lattice structure is comprised of interconnected struts, which, in principle, can be inclined at any angle (θ) from 0° to 90° , illustrated schematically in Fig. 1(a, b) through a body centered cubic (BCC) lattice unit (a) and a simple cubic lattice unit (b). The strut additive continuity model to be formulated below will encompass all potential inclination angles.

We first consider a theoretical lattice strut PBF process in which each layer of the metal powder will translate into the same layer thickness of dense solid (the shrinkage after melting and solidification will be discussed later). Fig. 2a illustrates such a PBF process for the manufacture of a strut of diameter d inclined at an angle of θ ($0^\circ < \theta \leq 90^\circ$). Each selectively melted cross section is assumed to be an ellipse, Fig. 2b, excluding the bottom and top layers within the build height of $d \times \cos\theta$. The symbol l_0 denotes the major axis of the last solidified ellipse, while l represents the overlap length between the solidified ellipses from two adjoining layers (n and $n+1$). The gap $\Delta l = l_0 - l$ defines the non-overlap region (Fig. 2b).

The following geometric relationships exist according to Fig. 2(a, b)

$$l_0 = \frac{d}{\sin\theta} \quad (1a)$$

$$l = \frac{d}{\sin\theta} - \frac{h_0}{\tan\theta} \quad (1b)$$

Clearly, for robust PBF of an inclined strut, l needs to be as close to l_0 as possible while the condition of $l = l_0$ occurs only when $\theta = 90^\circ$ or $\theta = 0^\circ$ (see Fig. 2c). The ratio of l to l_0 thus provides a measure of the degree of additive continuity for an inclined strut. Using φ to denote l/l_0 , we have

$$\varphi = 1 - \frac{h_0}{d} \times \cos\theta \quad (0^\circ < \theta \leq 90^\circ) \quad (2)$$

which varies between $1 - h_0/d$ (for $\theta \rightarrow 0^\circ$) and 1 (for $\theta \rightarrow 90^\circ$).

Eq. (2) did not consider the change in density from a powder-bed

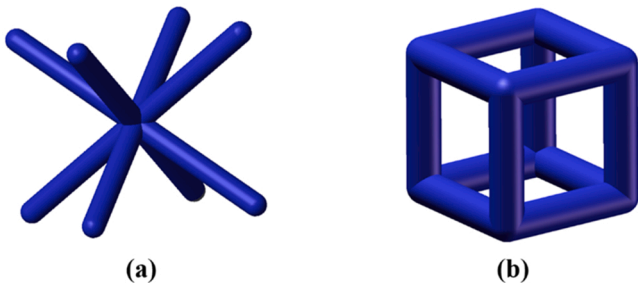


Fig. 1. Schematic illustration of inclined struts in a BCC lattice unit (a) and vertical and horizontal struts in a simple cubic lattice unit (b).

layer to molten metal and then to a dense solidified layer. In practice, after a thin layer of powder is melted, the resulting molten metal is denser than the powder layer. For Ti-6Al-4 V, the difference is $\sim 4.12 \text{ g cm}^{-3}$ for molten Ti-6Al-4 V [49] vs. $\sim 2.88 \text{ g cm}^{-3}$ for a layer of Ti-6Al-4 V powder in a typical powder bed, while the subsequent solidification shrinkage is $\sim 8 \text{ vol}\%$ [46]. Although limited to a thin layer, these shrinkage processes occur three dimensionally and are not uniform, thereby contributing to surface topography. However, it is noteworthy that the EB-PBF process of a lattice strut begins by forming its periphery or edge, followed by its interior. Once solidified, this thin periphery ($\sim 100 \mu\text{m}$) will not be remelted. In other words, the strut cross-sectional profile or diameter (d) is defined first, which is part of the EB-PBF AM strategy for quality build of a strut. Ignoring the shrinkage associated with the formation of this thin periphery, the 3D shrinkage is thus mainly restricted to the interior of each ellipse in Fig. 2b, which affects the layer thickness but not the periphery or cross-sectional size.

It has been found that for EB-PBF of Ti-6Al-4 V using the default parameters in Table 1, the resulting mean roughness (R_a) of the top horizontal surface is about $6 \mu\text{m}$ [47], equivalent to 12% of the layer thickness setting ($50 \mu\text{m}$). This gives an indication of the variation in layer thickness caused by the confluence of (i) the uneven melt flow, (ii) the two shrinkage processes just described, and (iii) other potential factors including vaporisation and melt spattering (ignoring any powder expansion on each hot solidified surface). Consequently, a mean layer thickness should always be used. For ease of description, we continue to use the symbol h below to denote this mean layer thickness.

The layer thickness evolves during a PBF process as follows. After the initial layer of powder with thickness h_0 is melted and solidifies, its thickness shrinks by $(1 - \rho_{\text{PBD}}/\rho_{\text{TD}})$. This increases the thickness of the 2nd layer to $h_0 + (1 - \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}})h_0$ as the build platform surface is kept at the same height. Similarly, the 3rd layer, after necessary mathematical rearrangements, has the thickness of

$$h_3 = h_0 + \left(1 - \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}}\right)h_0 + \left(1 - \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}}\right)^2 h_0 \quad (3)$$

The same trend continues and the thickness of the n th layer of powder (h_n) is defined by

$$h_n = h_0 \sum_{i=0}^{n-1} \left(1 - \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}}\right)^i \quad (4)$$

Solving the right-hand side of Eq. (4) yields

$$h_n = h_0 \left[1 - \left(1 - \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}}\right)^n\right] \times \frac{\rho_{\text{TD}}}{\rho_{\text{PBD}}} \quad (5)$$

Then replacing h in Eq. (2) with the actual layer thickness h_n we obtain

$$\varphi_n = 1 - \frac{h_0}{d} \times \cos\theta \times \left[1 - \left(1 - \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}}\right)^n\right] \times \frac{\rho_{\text{TD}}}{\rho_{\text{PBD}}} \quad (6)$$

Eq. (6), which reduces to Eq. (2) for $n = 1$, quantifies the dependence of lattice strut additive continuity on (i) the initial layer thickness setting h_0 , (ii) strut diameter d , (iii) strut inclination angle θ , (iv) powder-bed density ρ_{PBD} (affected by the powder-bed recoating process), and (v) layer number n . It provides a comprehensive description for PBF including μ -PBF [48].

In both Eq. (6) and Eq. (5), we have $\left(1 - \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}}\right)^n \rightarrow 0$ when $n \rightarrow \infty$ due to $0 < \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}} < 1$ but in reality, we have found that $(h_n - h_1)/h_1 < 10^{-3}$ always holds for $n \geq 10$ if $\rho_{\text{PBD}} \geq 0.5\rho_{\text{TD}}$. This enables us to define $n = 10$ as the cut-off value ($n_{\text{cut-off}}$) for PBF. This is because most PBF processes are expected to have $\rho_{\text{PBD}} \geq 0.5\rho_{\text{TD}}$, especially for PBF of intricate lattice structures, which requires $\rho_{\text{PBD}} \geq 0.6\rho_{\text{TD}}$ as identified subsequently. Eq. (6) then becomes

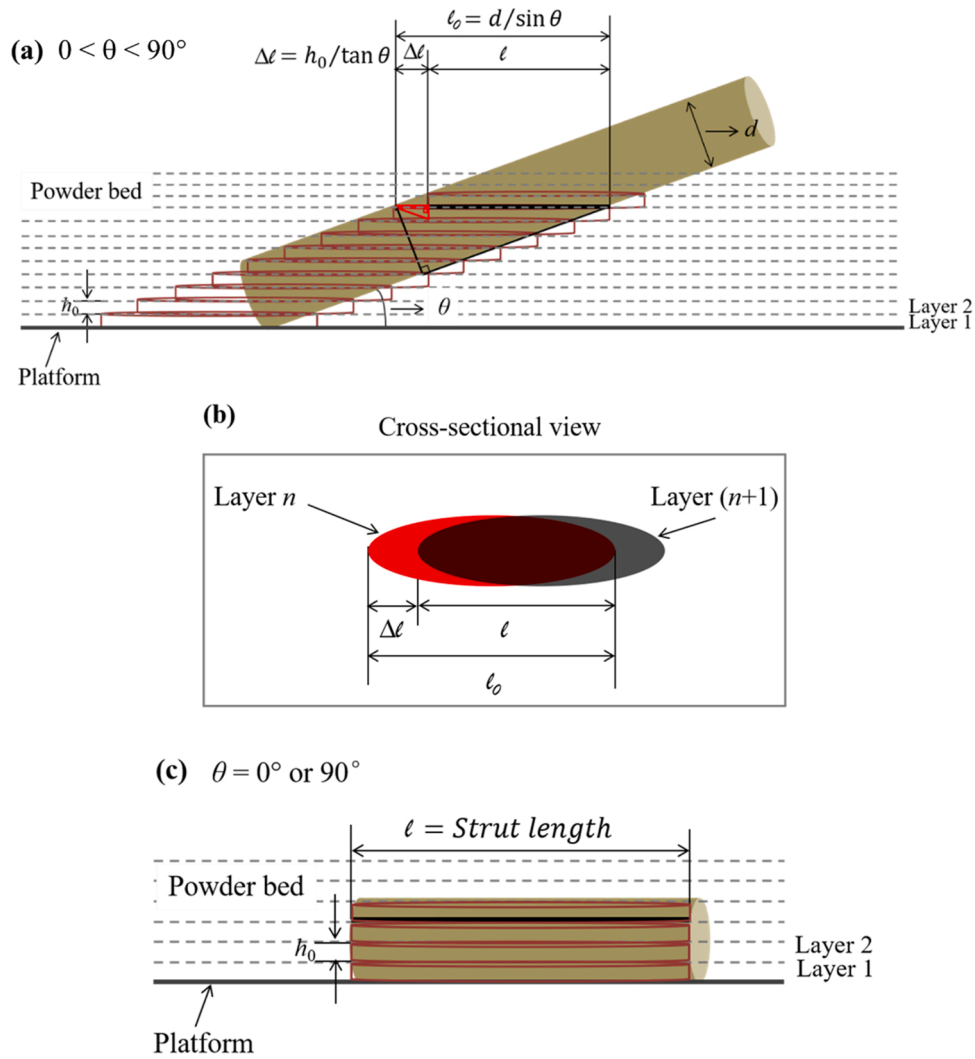


Fig. 2. (a) Illustration of PBF AM of a lattice strut inclined at $0^\circ < \theta \leq 90^\circ$. (b) A cross-sectional view of the inclined strut at layers n and $(n + 1)$. (c) PBF AM of a strut at $\theta = 0^\circ$ or 90° .

$$\varphi \approx 1 - \frac{h_0}{d} \times \cos \theta \times \frac{\rho_{TD}}{\rho_{PBD}} \quad (n \geq 10 \text{ and } \rho_{PBD} \geq 0.5\rho_{TD}) \quad (7)$$

Note that the selection of $n_{\text{cut-off}}$ depends on ρ_{PBD} . For example, $(h_n - h_5)/h_5 < 10^{-3}$ will hold for $n \geq 5$ rather than $n \geq 10$ if a dense powder bed such as $\rho_{PBD} = 0.75\rho_{TD}$ is used.

Fig. 3a depicts the dependence of h_n on layer number n by Eq. (5) with $h_0 = 50 \mu\text{m}$ and ρ_{PBD}/ρ_{TD} from 0.50 to 0.75. After an initial sharp rise, h_n stabilises at a value well above $h_0 = 50 \mu\text{m}$ in each case from a layer number between 5 and 10, depending on ρ_{PBD} . Spierings and Levy [49] once discussed the dependence of layer thickness on both n and ρ_{PBD} but to our knowledge Eq. (5) is the first rigorous description of h_n for PBF.

Corresponding to Fig. 3a, Fig. 3b shows the dependence of the strut additive continuity parameter φ on layer number n and ρ_{PBD} with $h_0 = 50 \mu\text{m}$, $d = 0.7 \text{ mm}$ and $\theta = 35^\circ$. Two observations are notable. One is that φ drops with increasing layer number n in the first 5–10 layers due to the sharp increase in layer thickness h_n (Fig. 3a). The other is that ρ_{PBD} displays a profound influence on φ due to its influence on h_n as well (Fig. 3a), where a higher ρ_{PBD} is always preferred. Note that ρ_{PBD} can be measured using a powder-bed capsule approach [50–52], which requires PBF of ~ 500 – 1000 layers. Alternatively, to a first approximation, the tap density of a powder (ρ_{TAP}) can be used to estimate ρ_{PBD} .

Other implications of Eq. (6) will be discussed via the subsequently constructed design maps. The next step is to determine a threshold value of φ for EB-PBF of uniform lattice struts and validate the link between the calculated quantities of φ and experimental observations of strut additive continuity.

3.2. Correspondence between φ and additive continuity of Ti-6Al-4 V lattice struts

We first designed and manufactured 165 Ti-6Al-4 V strut samples by EB-PBF with an initial layer thickness setting of $h_0 = 50 \mu\text{m}$. A review of the literature on EB-PBF of Ti-6Al-4 V lattices has indicated that the strut diameter designed for basic research typically falls in the range of 0.1–1.0 mm, inclined from 0 to 90° [8]. However, for end applications that require high compressive strength, thicker struts up to 3.0 mm in diameter are often needed according to our experience. For these reasons, the strut diameters considered in this study include 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.5, 2.0, 2.5, 3.0 mm, inclined at 90, 75, 60, 45, 35, 25, 20, 15, 10, 5, 0° . The minimum strut diameter 0.05 mm, which is below both the Dv(10) (52 μm) of the Ti-6Al-4 V powder and the EB size used (Table 1), is chosen to test the limit of the EB-PBF system employed. These combinations are expected to have covered most practical cases in lattice design for PBF. [Supplementary](#)

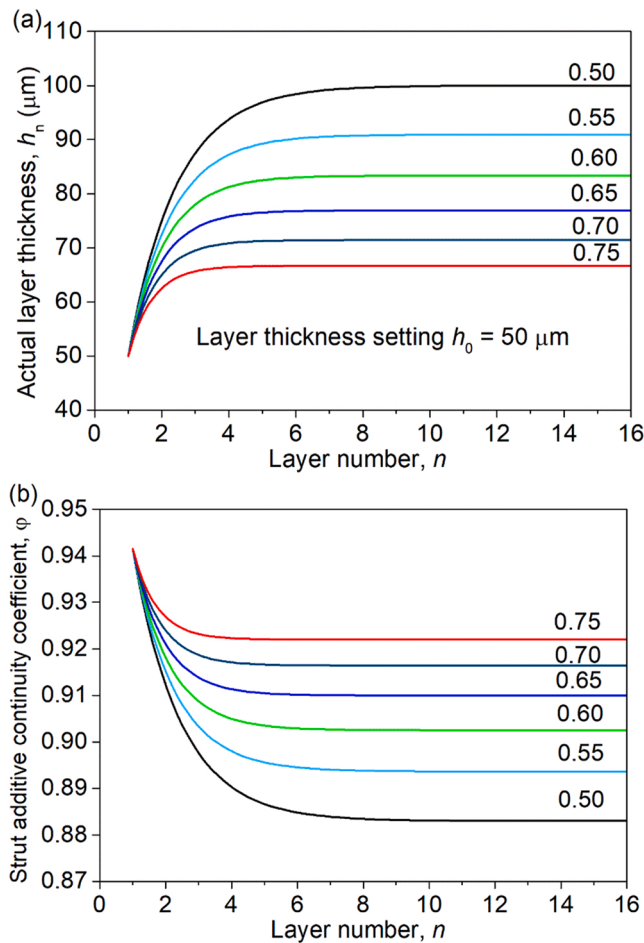


Fig. 3. (a) Actual mean layer thickness h and (b) strut additive continuity parameter ϕ vs. layer number n and powder-bed relative density $\rho_{\text{PBD}}/\rho_{\text{TD}}$ (figures on lines) for PBF of a 35°-inclined lattice strut of 0.7 mm in diameter at the initial layer thickness setting of 50 μm .

Fig. S1 shows a view of all the manufactured Ti-6Al-4 V lattice strut samples.

The value of ϕ of each manufactured strut was calculated from Eq. (7) by taking the measured $\rho_{\text{TAP}} = 2.88 \text{ g cm}^{-3}$ as the approximate ρ_{PBD} , which gives $\rho_{\text{PBD}}/\rho_{\text{TD}} = 0.65$ ($\rho_{\text{TD}} = 4.42 \text{ g cm}^{-3}$ for Ti-6Al-4 V). Since not all manufactured lattice struts are uniform, the value of ϕ varies along the axis of a non-uniform strut and reaches the minimum (ϕ_{min}) at its thinnest section according to Eq. (7). Therefore, we measured the diameter of each strut at its thinnest section and calculated ϕ_{min} . The results are listed in **Supplementary Table S1** for all the 165 strut samples. **Fig. 4** exemplifies the correspondence between the value of ϕ_{min} and the strut morphology for 39 lattice strut samples manufactured at low inclines of 5°, 10°, 15° and 35° (low inclines are more challenging to manufacture). An approximately consistent borderline value of ϕ_{min} , in the narrow range of 0.89–0.91, is observed for each group of strut samples (see the red broken lines in **Fig. 4a–d**), above which lattice struts tend to be uniform while below which noticeable variations exist along the strut axis (see the red arrows in **Fig. 4a–d**). This observation applies to all the lattice strut samples in **Fig. S1**. Accordingly, we choose $\phi_{\text{min-threshold}} = 0.90$ as a provisional minimum threshold for EB-PBF of uniform Ti-6Al-4 V lattice struts. Its applicability to lattice design and EB-PBF AM will be assessed below in detail through selected lattice structures.

It should be accentuated that the strut additive continuity results shown in **Fig. 4** and **Table S1** were obtained from using the Arcam default parameters in **Table 1**, which allow for both a high build rate

(60–80 cm^3/h) and a nearly pore-free density ($> 99.92\%$ by $\mu\text{-CT}$ even for build heights up to 300 mm [44]). Hence, they are a set of almost ideal EB-PBF parameters for Ti-6Al-4 V. The use of other parameters is likely to deteriorate the additive continuity, leading to more lattice struts resembling those with ϕ clearly below 0.90 in **Fig. 4**. The selection of ϕ and factors affecting its value including EB-PBF variables will be discussed in **Section 4.3**.

3.3. The $\phi_{\text{min-threshold}}$ -informed lattice design and validation

Tetrahedron and octahedron lattice structures, which contain struts inclined at 35°, 45° and 60°, are selected and designed with and without respecting $\phi_{\text{min-threshold}} = 0.90$ for EB-PBF with Ti-6Al-4 V for systematic evaluation. Three scenarios are considered, namely (i) $\phi < 0.90$, (ii) $\phi = 0.90$ and (iii) $\phi > 0.90$. The results are described below.

Scenario 1 ($\phi < 0.90$) **Fig. 5a** shows a tetrahedron unit cell that comprises horizontal (0°, orange), vertical (90°, green), 35°-inclined (blue) and 45°-inclined (red) struts. We first designed a tetrahedron lattice sample with dimensions of $58 \times 58 \times 87 \text{ mm}^3$ (height: 87 mm) by experience without considering the $\phi_{\text{min-threshold}}$ criterion. The lattice sample contains 3000 35°-inclined and 3300 45°-inclined struts (strut diameter: 0.3 mm, length: 5 mm). **Fig. 5c** shows the manufactured design in Ti-6Al-4 V using default EB-PBF parameters (**Table 1**, $h_0 = 50 \mu\text{m}$). At first glance, our empirical design was successful. However, a closer inspection revealed severe structural failures. As shown in **Fig. 5** (d, e), both the 35°- and 45°-inclined struts exhibited barely connected necking sections (see red arrows). We examined all the 1200 struts that are inclined at 35° and 45° on the six external surfaces of the lattice sample and identified that nearly a third of them displayed similar structural faults. This lattice design is thus a failure, despite the use of a well-established EB-PBF process and powder feedstock. The as-manufactured strut diameters are 0.31 mm for the 35°-inclined struts and 0.32 mm for the 45°-inclined ones. Using Eq. (7), we obtain $\phi = 0.797$ for the 35°-inclined and $\phi = 0.830$ for the 45°-inclined struts, both of which are unsurprisingly well below $\phi_{\text{min-threshold}} = 0.90$.

Scenario 2 ($\phi = 0.90$) The above tetrahedron lattice was re-designed as per $\phi_{\text{min-threshold}} = 0.9$, **Fig. 6a**, which still contains 3000 35°-inclined and 3300 45°-inclined struts but measures $27 \times 27 \times 40 \text{ mm}^3$ (height: 40 mm) in dimensions. To ensure $\phi = 0.90$ for each inclined strut, we increased the strut diameter to 0.61 mm (3 mm long) from 0.3 mm according to Eq. (7). The EB-PBF process was identical (**Table 1**). The as-manufactured lattice is shown in **Fig. 6b**. We examined all the 600 inclined struts on the six external surfaces of the lattice sample. No severe necking or structural faults were observed. All the inclined struts displayed an approximately uniform profile along their axial direction, **Fig. 6c**, demonstrating a substantial improvement in quality. **Fig. 6d** shows a representative view of the 35°-inclined struts. The strut shows a well-built profile, where the rough surface is due to the high magnification of inspection. Furthermore, to assess the additive continuity of all the 5700 inclined struts (both 35°- and 45°-inclined) inside the lattice, we scanned the entire lattice with $\mu\text{-CT}$ (Methods). Internal defects in the struts as small as 30 μm were detected (the dark spots in **Fig. 6f**) but no discontinuity or severe necking was observed in each inclined strut. Rather, as revealed in **Fig. 6f**, robust inclined struts were manufactured.

Scenario 3 ($\phi > 0.90$) To further test our $\phi_{\text{min-threshold}} = 0.90$ criterion, we designed and manufactured a truncated octahedron Ti-6Al-4 V lattice using EB-PBF with a high ϕ value (0.964) by keeping $h_0 = 50 \mu\text{m}$. **Fig. 7a** shows a unit cell of the truncated octahedron lattice (orange: horizontal, 0°; blue: 60°-inclined). The lattice design ($48 \times 48 \times 48 \text{ mm}^3$) is displayed in **Fig. 7b**, which contains 16,800 60°-inclined struts with a design diameter of 1.06 mm. This results in $\phi = 0.964$ by Eq. (7). **Fig. 7c** shows the manufactured lattice structure. A representative view of the 60°-inclined struts is presented in **Fig. 7d**. All the 1600 struts inclined at 60° on the six external surfaces exhibited outstanding additive continuity with high uniformity and consistency,

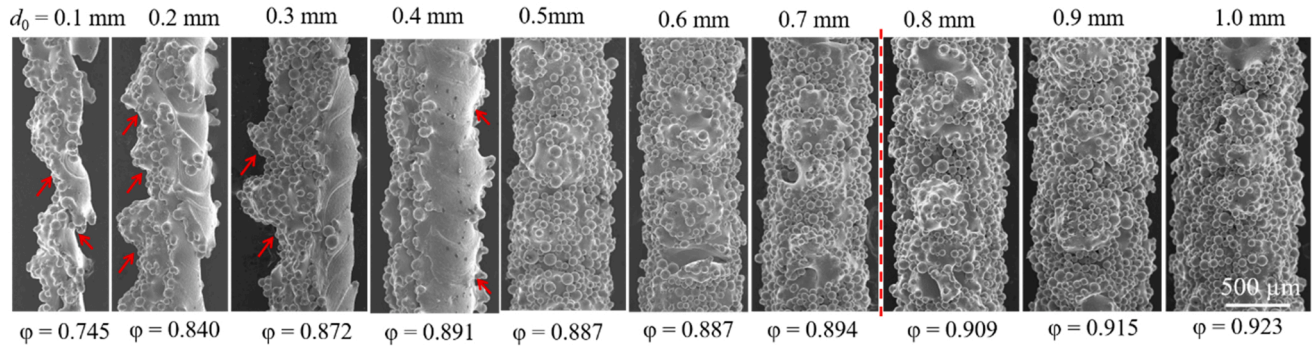
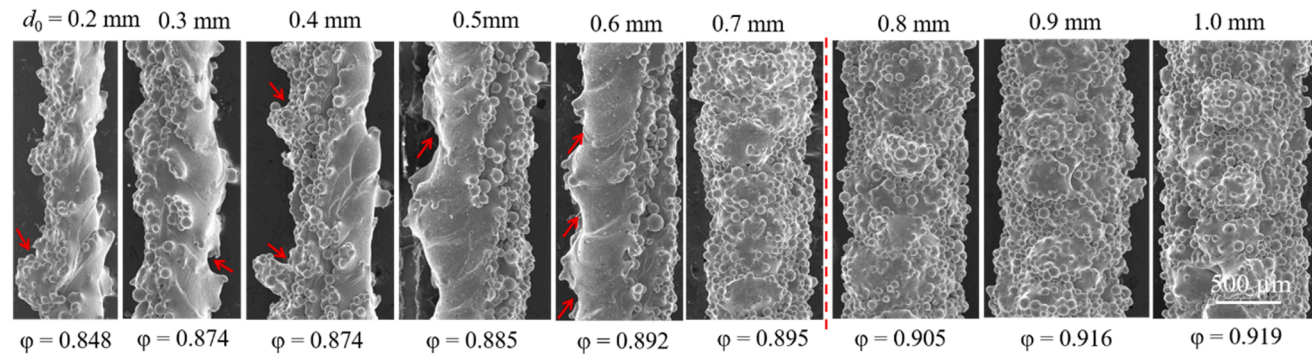
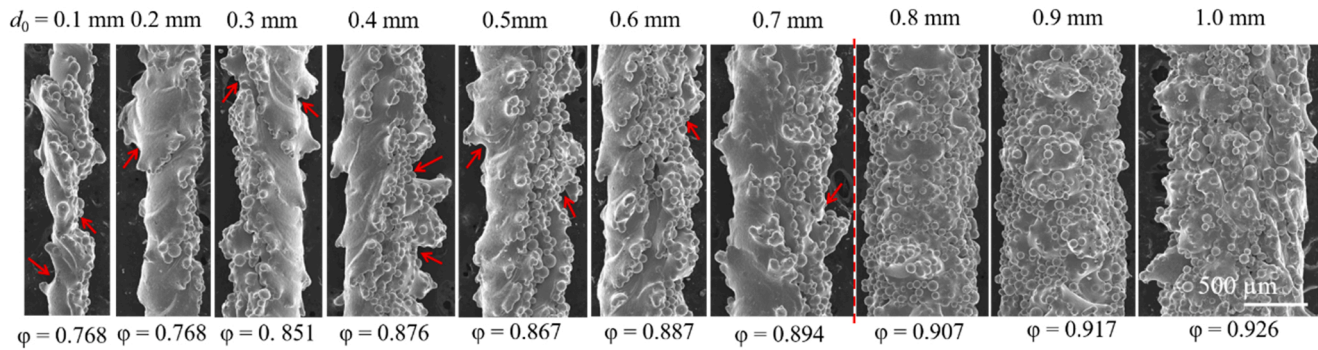
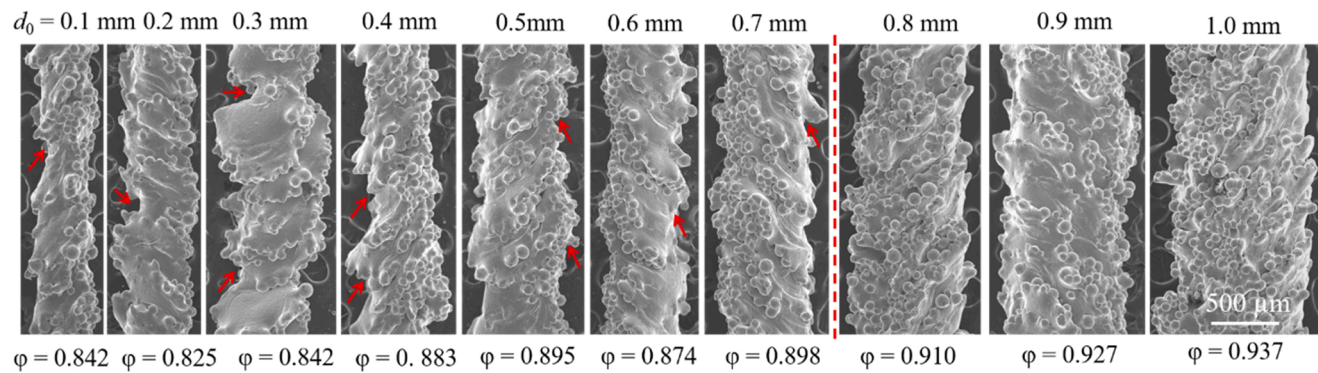
(a) $\theta = 5^\circ$ **(b) $\theta = 10^\circ$** **(c) $\theta = 15^\circ$** **(d) $\theta = 35^\circ$** 

Fig. 4. Correspondence between $\phi_{\min}$ and strut morphology for Ti-6Al-4 V struts manufactured by EB-PBF at (a) $\theta = 5^\circ$; (b) $\theta = 10^\circ$; (c) $\theta = 15^\circ$; and (d) $\theta = 35^\circ$. In each group, strut design diameters are increased at an interval of 0.1 mm from 0.1 to 1.0 mm. Uniform struts are ensured with $d > 1.0 \text{ mm}$ due to their high values of $\phi_{\min}$ and therefore not included.

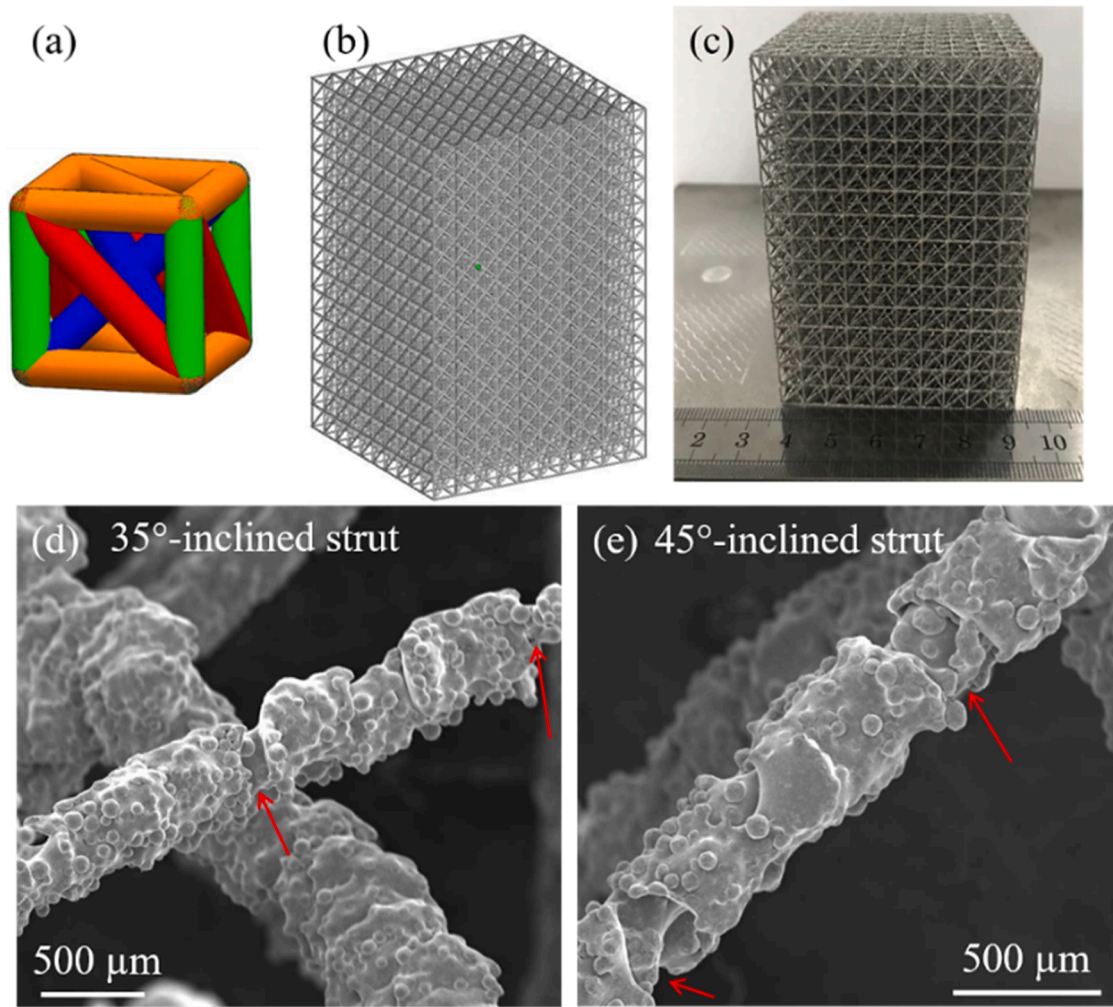


Fig. 5. Ti-6Al-4 V tetrahedron lattice by EB-PBF: (a) a tetrahedron unit with horizontal (0° , orange), vertical (90° , green), 35° -inclined (blue), and 45° -inclined (red) struts; (b) an empirical tetrahedron lattice design ($58 \times 58 \times 87 \text{ mm}^3$); (c) manufactured lattice; and (d, e) closer views of the inclined struts from (c) showing clear structural faults due to $\varphi < \varphi_{\text{min-threshold}} = 0.90$.

akin to those shown in Fig. 7d.

3.4. Strut additive continuity maps for PBF of Ti-6Al-4 V and other lattice materials

The establishment of Eqs. (6) and (7) and $\varphi_{\text{min-threshold}} = 0.90$ allows us to map out comprehensive design guidance for PBF of metal lattices for any specified powder feedstock. The resulting design maps are shown in Fig. 8 for $\rho_{\text{PBD}} = 0.55\rho_{\text{TD}}$, Fig. 9 for $\rho_{\text{PBD}} = 0.65\rho_{\text{TD}}$ (for EB-PBF of Ti-6Al-4 V in this research), and Fig. 10 for $\rho_{\text{PBD}} = 0.75\rho_{\text{TD}}$. More scenarios are shown in Figs. S3–S7. Together they cover most practical cases with h_0 ranging from $20 \mu\text{m}$ to $100 \mu\text{m}$ and ρ_{PBD} from $0.50\rho_{\text{TD}}$ to $0.75\rho_{\text{TD}}$. The experimental data obtained from the 165 Ti-6Al-4 V strut samples with $h_0 = 50 \mu\text{m}$ and $\rho_{\text{PBD}} = 0.65\rho_{\text{TD}}$ is superimposed on Fig. 9c. High consistency was observed. Their implications will be discussed below.

4. Discussion

4.1. Initial layer thickness setting h_0

As observed from Figs. 8–10 and Figs. S3–S7, the initial layer thickness setting (h_0) profoundly affects lattice strut additive continuity for each given powder-bed density ρ_{PBD} . The smaller the h_0 , the broader the green region (robust additive continuity) in each case examined. For

example, at $h_0 = 20 \mu\text{m}$, additive continuity is a concern only for lattice struts of thinner than $310 \mu\text{m}$ in the case of Ti-6Al-4 V with $\rho_{\text{PBD}} = 0.65\rho_{\text{TD}}$ (Fig. 9a), irrespective of strut orientation. If h_0 is further reduced to $10 \mu\text{m}$, then all Ti-6Al-4 V struts that are thicker than $155 \mu\text{m}$ can be robustly manufactured for all inclination angles according to Fig. S3. Thin-layer PBF ($< 20 \mu\text{m}$) is thus ideal for intricate lattices. However, this will necessitate the use of fine powder, whose strong cohesiveness can easily hinder powder spreading [53,54]. Furthermore, the increased cost of production for fine spherical powders ($\leq 20 \mu\text{m}$) and their safe handling and transportation can be significant hindrances as well [48]. As such, thin-layer PBF has been less commonly used except for μ -PBF [48].

Technically, one can always choose to use a small initial layer thickness h_0 ($\leq 20 \mu\text{m}$) for a selected powder feedstock but this can cause several consequences. For example, the particle size distribution (PSD) of the commonly used $15\text{--}45 \mu\text{m}$ Ti-6Al-4 V powder from GE AP&C for L-PBF has $D_v(10) = 20 \mu\text{m}$ and $D_v(50) = 34 \mu\text{m}$ [55]. If one chooses $h_0 = 20 \mu\text{m}$ for this powder, then more than 50 vol% of the powder will be pushed out of the build platform by the recoating blade, because the stabilised layer thickness $h_{10} = 31 \mu\text{m}$ defined by Eq. (5) is less than $D_v(50)$. Not only will this lead to low powder utilisation, powder contamination and extra powder rehandling but also low ρ_{PBD} due to the radically changed PSD, which often corresponds to more defects. Future development may allow more flexibility but the current

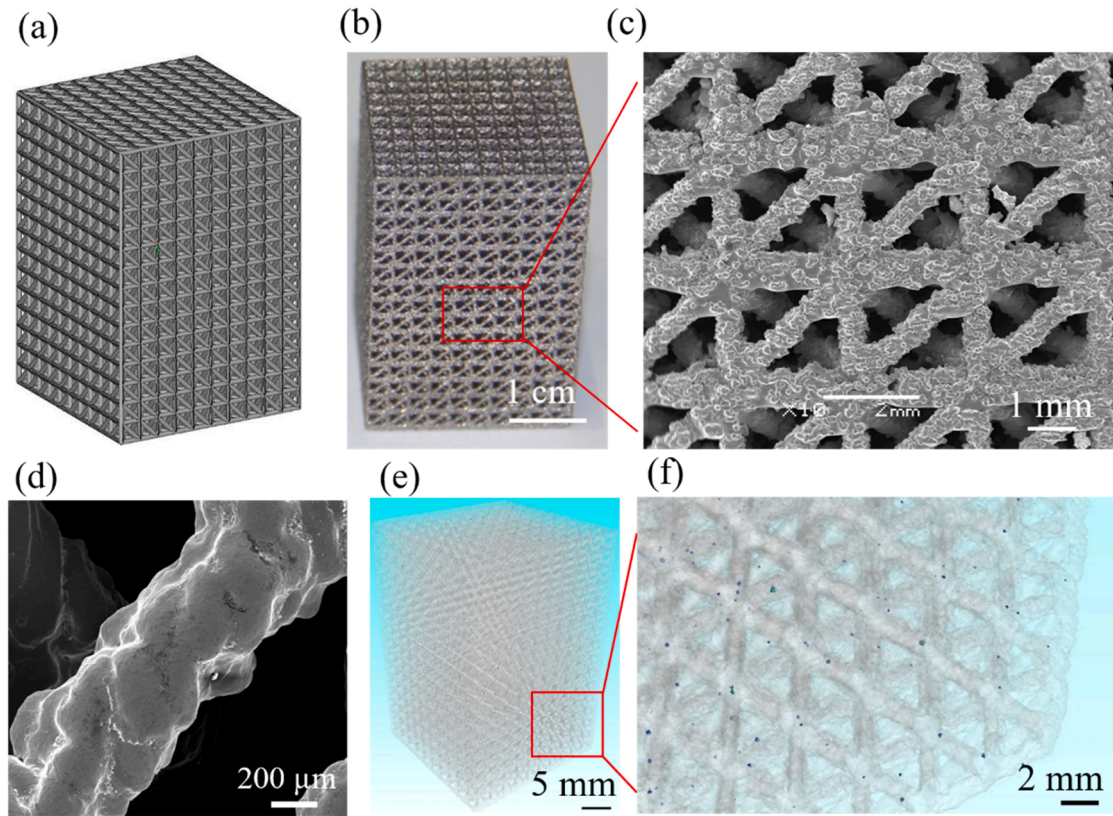


Fig. 6. (a) Re-designed tetrahedron lattice ($27 \times 27 \times 40 \text{ mm}^3$) as per $\varphi_{\text{min-threshold}} = 0.90$; (b, c) the manufactured lattice in Ti-6Al-4 V; (d) SEM view of a 35° -inclined strut in (b); (e) μ -CT scan of the entire lattice; and (f) closer μ -CT views of the tetrahedron lattice. No severe necking or structural faults were detected in the inclined struts inspected even though internal defects as small as $30 \mu\text{m}$ were observed (see the dark spots in Fig. 6f).

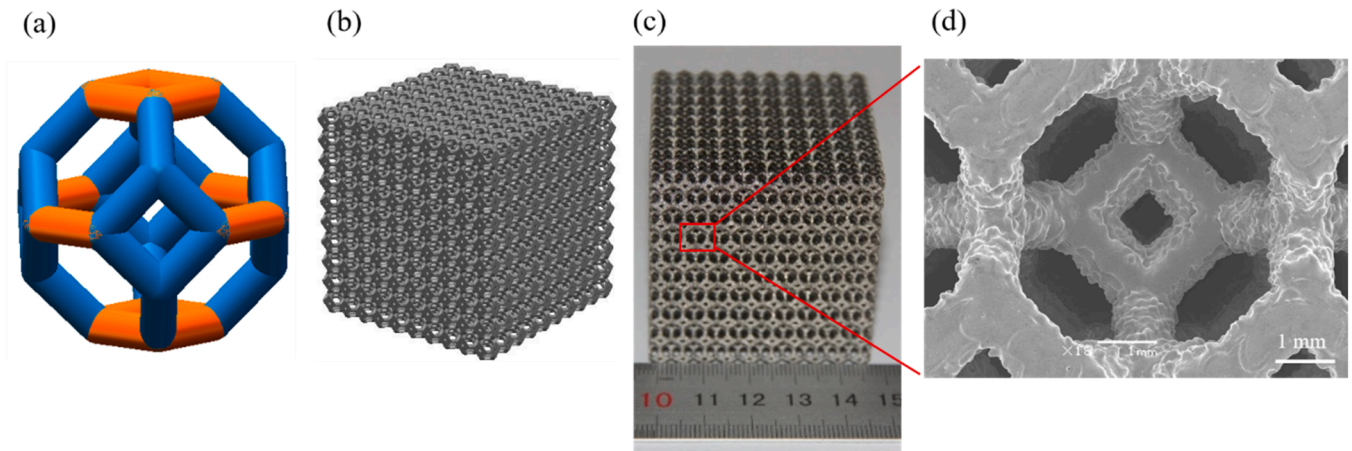


Fig. 7. Truncated octahedron lattice of Ti-6Al-4 V by EB-PBF: (a) unit cell (orange: horizontal, 0° ; blue: 60° -inclined); (b) lattice design with $\varphi = 0.964$; (c) manufactured lattice ($48 \times 48 \times 48 \text{ mm}^3$) containing 16,800 struts inclined at 60° ; and (d) 60° -inclined lattice struts from (c).

mainstream PBF practice, namely $30\text{--}40 \mu\text{m}$ for L-PBF and $50\text{--}70 \mu\text{m}$ for EB-PBF, is expected to continue for PBF AM of most metal lattice materials.

4.2. The impact of powder-bed density ρ_{PBD} and PSD

The important influence of ρ_{PBD} on lattice strut additive continuity can be inferred at the outset from Fig. 3b (a 35° -inclined strut with $h_0 = 50 \mu\text{m}$). Figs. 8–10 and Figs. S3–S7 allow for a comprehensive assessment of the influence of ρ_{PBD} from $0.5\rho_{\text{TD}}$ to $0.75\rho_{\text{TD}}$ with respect to $h_0 = 20\text{--}100 \mu\text{m}$ and $\theta = 0\text{--}90^\circ$. A notable finding is that if a powder

feedstock exhibits $\rho_{\text{PBD}} < 0.6\rho_{\text{TD}}$, then it becomes challenging to satisfy $\varphi_{\text{min-threshold}} = 0.90$ for $\theta \leq 45^\circ$, $d < 0.8 \text{ mm}$ and $h_0 = 50 \mu\text{m}$ or for $\theta \leq 45^\circ$, $d < 0.5 \text{ mm}$ $h_0 = 30 \mu\text{m}$. The former prediction concurs with the systematic observations in Fig. 4, where the borderline strut diameter consistently sits at 0.8 mm for $\theta = 5^\circ, 10^\circ, 15^\circ, 35^\circ$ with $\rho_{\text{PBD}} = 0.65\rho_{\text{TD}}$ (note that reducing ρ_{PBD} from $0.65\rho_{\text{TD}}$ to $< 0.60\rho_{\text{TD}}$ will increase this borderline diameter to greater than 0.8 mm). Therefore, we propose that the powder selection criteria for intricate lattices should consider including a new important requirement of $\rho_{\text{PBD}} \geq 0.60\rho_{\text{TD}}$ (the powder recoating system affects ρ_{PBD}). This differs from PBF of non-lattice materials and could potentially exclude certain metal or alloy

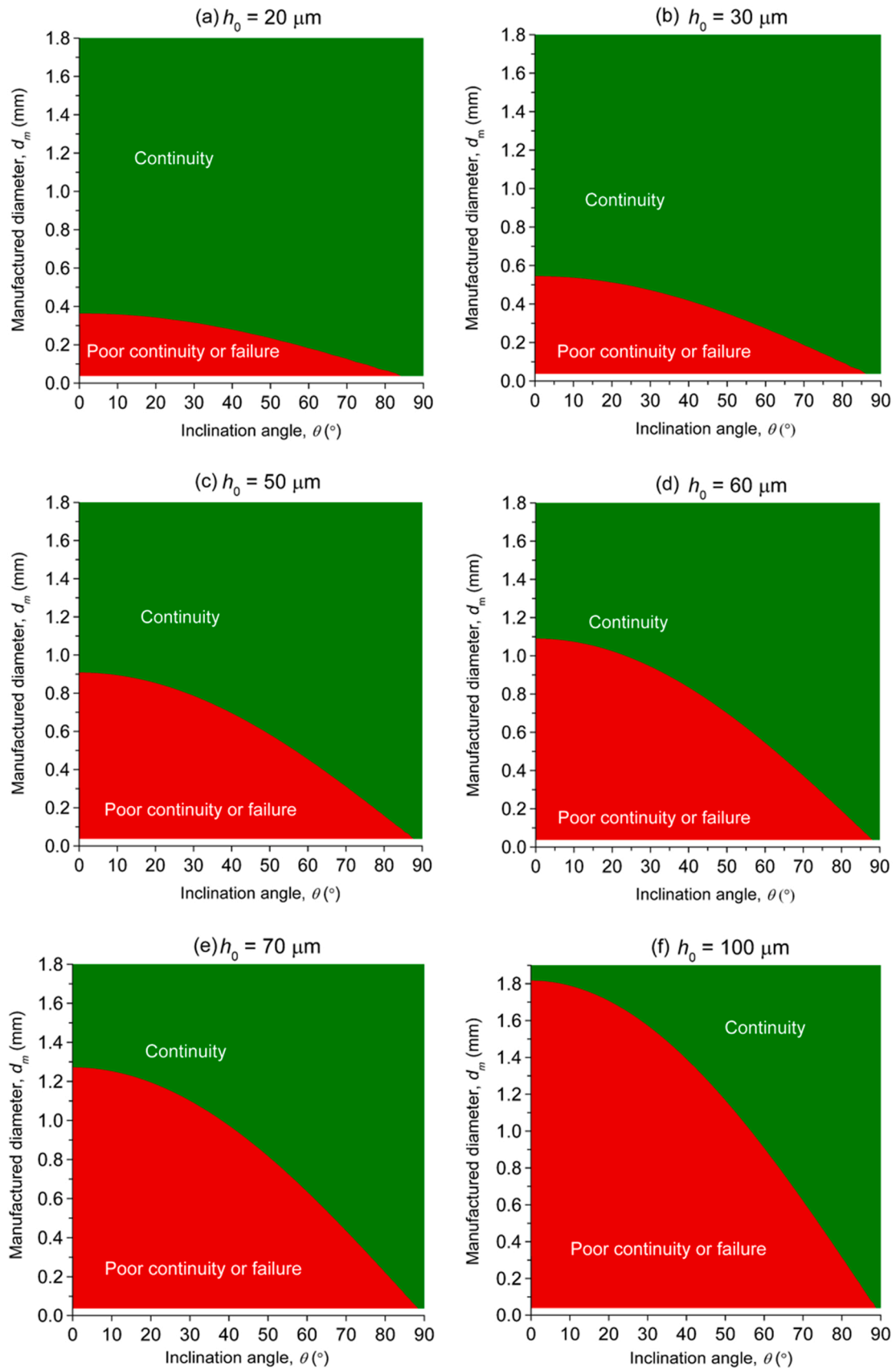


Fig. 8. Strut additive continuity maps for PBF of metal lattices with $\rho_{\text{PBD}} = 0.55\rho_{\text{TD}}$.

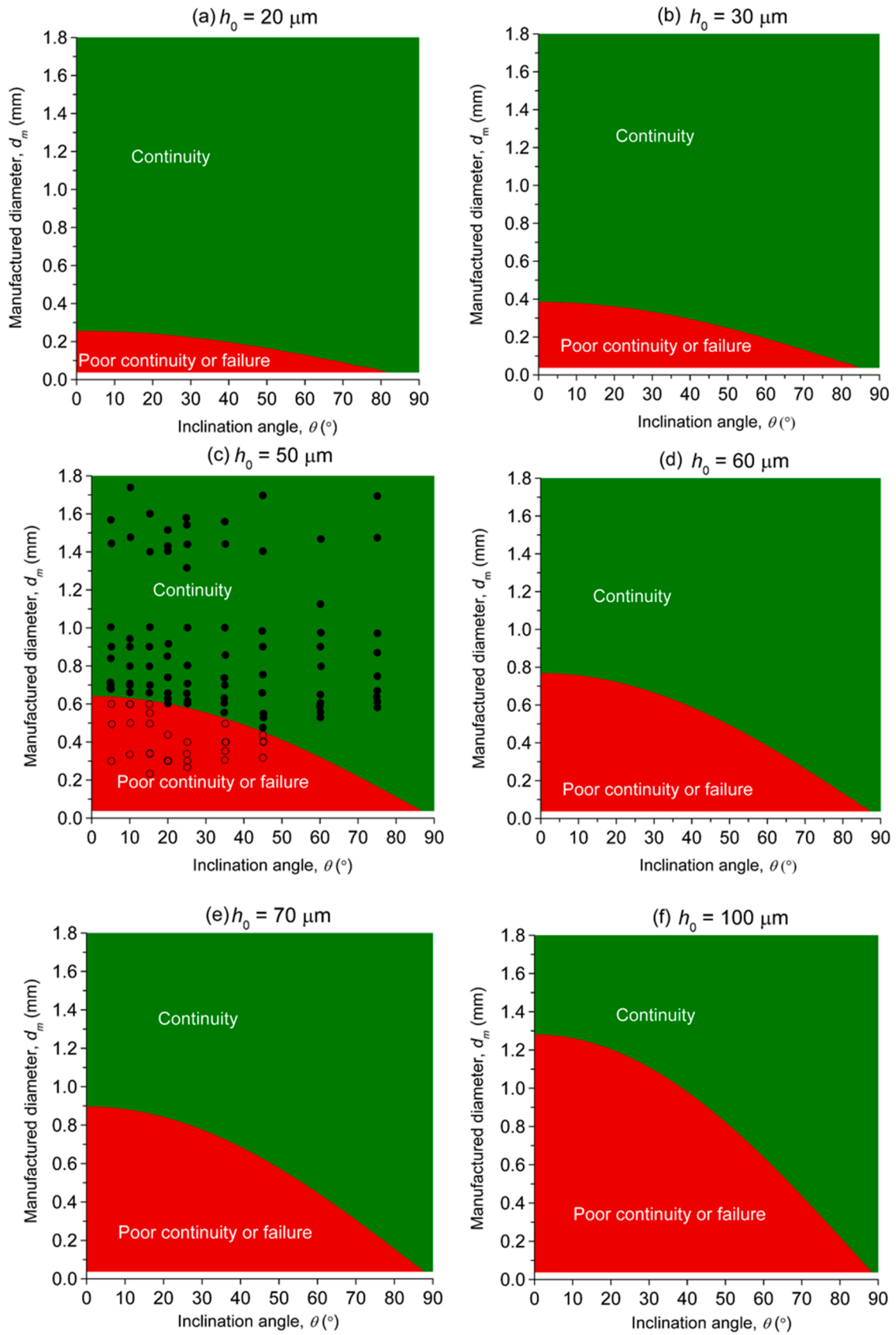


Fig. 9. Strut additive continuity maps for PBF of Ti-6Al-4 V lattices ($\rho_{\text{PBD}} = 0.65\rho_{\text{TD}}$). The data obtained from the 165 strut samples with $h_0 = 50 \mu\text{m}$ is superimposed on (c).

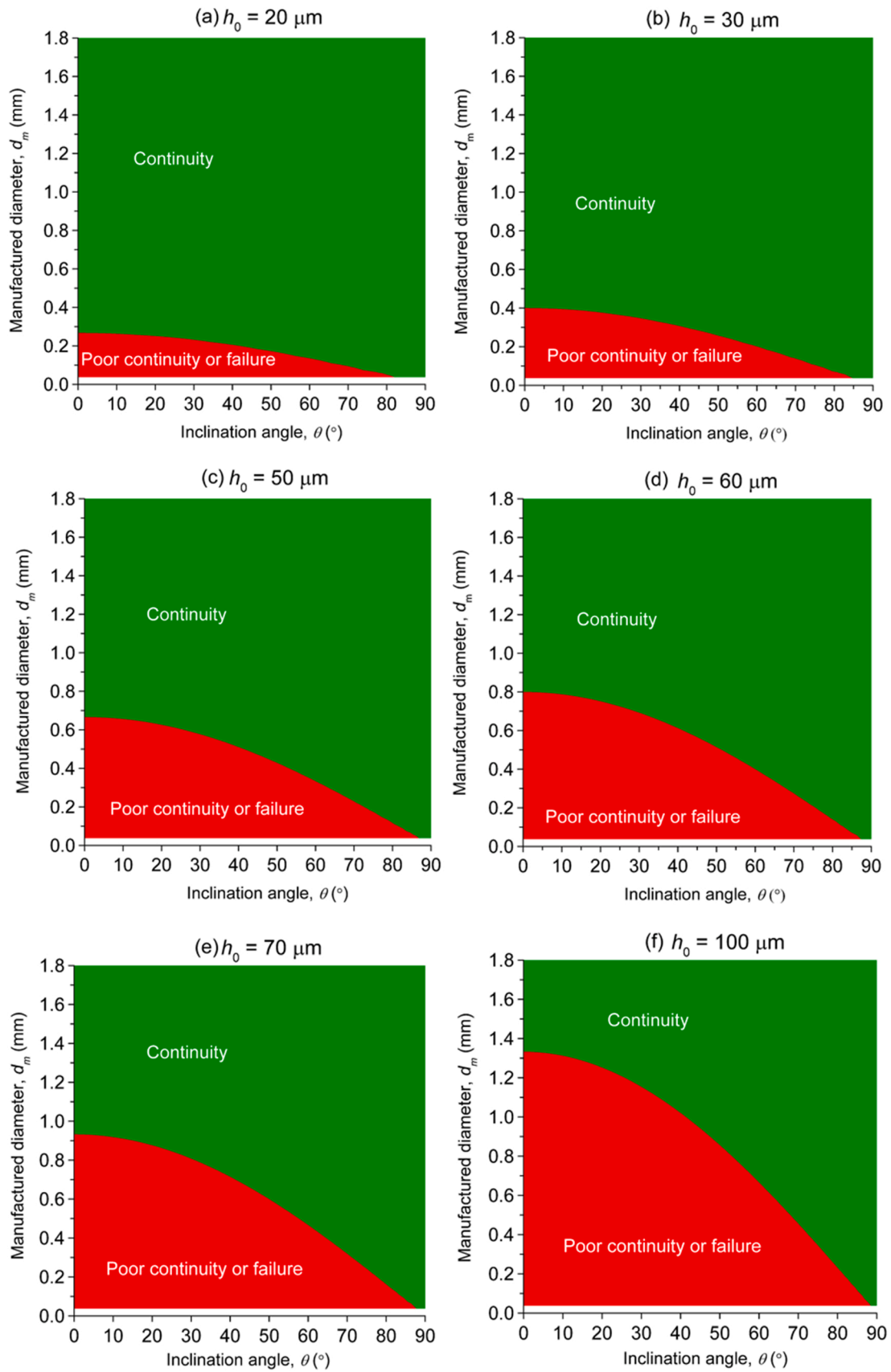


Fig. 10. Strut additive continuity maps for EB-PBF of metal lattices with $\rho_{\text{PBD}} = 0.75\rho_{\text{TD}}$.

powder products.

The ρ_{PBD} of a spherical powder depends on its PSD [49–52,56–59]. The following relationships have been proposed for both effective use and packing of the powder for PBF [49]:

$$\frac{h_{\text{actual}}}{Dv(90)} = 1.5 \quad (8)$$

$$\frac{Dv(90)}{Dv(10)} = 5 \quad (9)$$

where h_{actual} is the actual layer thickness, corresponding to the stabilised layer thickness h_{10} identified earlier with Eq. (5). Eq. (8) allows accommodation of almost all the powder particles within the thickness of each stabilised layer, while Eq. (9) is for effective filling between coarse particles (multimodal PSD leads to higher ρ_{PBD} [56–59]). Substituting $h_{\text{actual}} = 1.5Dv(90)$ into Eq. (5) as h_{10} by noting that $(1 - \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}})^n \rightarrow 0$ when $n \geq 10$ for $\rho_{\text{PBD}} \geq 0.5\rho_{\text{TD}}$, we have

$$h_0 = 1.5Dv(90)\frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}} \quad (10)$$

In principle, the implementation of Eq. (10) can ensure high utilisation of the powder. However, it will result in a high initial layer thickness setting of h_0 . For example, the widely used Ti-6Al-4 V powder products from GE AP&C have $Dv(90) = 44, 54$ or $61 \mu\text{m}$ for L-PBF and $Dv(90) = 102$ or $123 \mu\text{m}$ for EB-PBF [55]. The application of Eq. (10) will lead to $h_0 = 43\text{--}60 \mu\text{m}$ for L-PBF and $h_0 = 100\text{--}120 \mu\text{m}$ for EB-PBF ($\rho_{\text{PBD}} = 0.65\rho_{\text{TD}}$). As is clear from Figs. 8–10 and Figs. S3–S7, such settings of h_0 will substantially reduce the design space.

To limit the amount of powder that is pushed out of the powder bed to ≤ 10 vol% while ensuring a flexible design space and a high ρ_{PBD} (i.e., $\geq 0.6\rho_{\text{TD}}$), the use of a workable thin layer tends to increase ρ_{PBD} [34], we propose that $Dv(90)$, rather than $1.5 Dv(90)$, be set as the stabilised layer thickness. This reduces Eq. (10) to

$$h_{0\text{-specific}} = Dv(90)\frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}} \quad (11)$$

This provides a powder-specific baseline initial layer thickness setting for PBF AM of lattice materials, which can be taken as $\sim 0.6Dv(90)$. It could be considered for PBF AM of non-lattice products as well.

4.3. The selection of φ and effect of EB-PBF processing variables

The value of φ varies from 0 and 1.0, where $\varphi = 1.0$ corresponds to vertical struts ($\theta = 90^\circ$) while $\varphi = 0$ corresponds to $l = 0$ in Fig. 2b, i.e., a complete failure with no overlap occurring between two ellipses. The objective is always to achieve $\varphi \rightarrow 1.0$ where practical. There exist only two options by Eq. (7). One is to increase θ to the vicinity of 90° to let $\cos\theta \rightarrow 0$, which means to virtually eliminate all inclined struts (impractical). The other is to make $h_0/d < 1$, which is more practical. We will focus on this option. Note that for EB-PBF of Ti-6Al-4 V with the Arcam powder, the default layer thickness setting of $h_0 = 50 \mu\text{m}$ has already approached the lower limit. This is because the powder has $Dv(10) = 52 \mu\text{m}$ and $Dv(50) = 71 \mu\text{m}$, while the stabilized layer thickness h is only $77 \mu\text{m}$ by Eq. (5) ($\rho_{\text{PBD}} = 0.65\rho_{\text{TD}}$). Consequently, about half of the powder will be pushed out of the powder bed, which is already a significant penalty for choosing $h_0 = 50 \mu\text{m}$. Therefore, it is only realistic to make $h_0/d < 1$ through increasing strut diameter d .

Table 2 lists the minimum strut diameter d_{min} required for each selected value of φ by also considering the influence of strut inclination angle. While the objective is to have $\varphi \rightarrow 1$, the reality is that this condition requires the use of unusually thick struts. For example, the selection of $\varphi = 0.999$ requires $d_{\text{min}} = 38.5 \text{ mm}$ for $\theta > 60^\circ$ and $d_{\text{min}} = 75.75 \text{ mm}$ for $\theta > 10^\circ$. Even for the selection of $\varphi = 0.99$, it still requires $d_{\text{min}} = 3.85 \text{ mm}$ for $\theta > 60^\circ$ and $d_{\text{min}} = 7.58 \text{ mm}$ for $\theta > 10^\circ$. This clearly diverges from the design practice of intricate lattice materials

Table 2

Minimum strut diameter d_{min} versus φ and θ by Eq. (7).

φ	Minimum strut diameter d_{min} (mm) for selected φ ($h_0 = 50 \mu\text{m}$; $\rho_{\text{PBD}} = 0.65\rho_{\text{TD}}$)		
	$10^\circ \leq \theta \leq 30^\circ$	$30^\circ < \theta \leq 60^\circ$	$60^\circ < \theta < 90^\circ$
0.900	0.758	0.666	0.385
0.925	1.010	0.888	0.513
0.950	1.515	1.332	0.769
0.990	7.575	6.662	3.846
0.999	75.754	66.617	38.462

where d is often less than or in the vicinity of 1 mm. As a result, a compromise is necessary between strut additive continuity and design flexibility. The selection of $\varphi_{\text{min-threshold}} = 0.90$ is one such compromise, based on experimental observations, which requires $d_{\text{min}} = 0.39$ for $\theta > 60^\circ$ and $d_{\text{min}} = 0.76 \text{ mm}$ for $\theta > 10^\circ$ (Table 2), converging with the past and present design practice. The selection of $\varphi = 0.925$ or 0.950 will certainly enhance the strut additive continuity but will squeeze the design space, i.e., the green zones in Figs. 8–10 and Figs. S3–S7, due to the increased d_{min} (Table 2). The condition of $\varphi_{\text{min-threshold}} = 0.90$, as can be inferred from Fig. 2b ($l/l_0 = 0.9$), may be visualized in such a way that the additive continuity of each strut achieves 90% of its ideal case ($\varphi = 1.0$) while there is still significant design flexibility.

The two types of severe geometric defects when φ is clearly below 0.90 are the increased presence of narrow sections along the strut axis and protrusions with or without powder adherence. Both defects are rich in the struts with $\varphi < 0.88$ in Fig. 4(a–d). This is not surprising by the φ -model. Referring to Fig. 2b, what is changing physically behind the selected threshold is that the smaller the value of φ , the larger the non-overlapping length or uncovered staircase length Δl will be, and therefore the easier the formation of narrow sections. Furthermore, this uncovered staircase length Δl serves as a natural base for powder adherence, visible in Fig. 4a (0.2, 0.3), Fig. 4b (0.2, 0.4) and other images, which is highly undesired.

Aside from the strut geometric design considerations, powder characteristics and layer thickness, EB-PBF processing conditions can also affect strut additive continuity. For example, increasing the layer dwell time before next recoating can allow each as-solidified strut cross section to cool down to a sufficiently low temperature such that powder adherence on the Δl region in Fig. 2b can be mitigated or even avoided. Furthermore, research has shown that ensuring a steady shallow elliptical melt pool is essential to produce smooth surfaces by PBF (down to $R_a = 3\text{--}5 \mu\text{m}$ by normal L-PBF) [60]. This was realized through using a large beam size and a low volumetric energy density E_v [60], by considering the influence of the major PBF processing variables, namely beam power, beam size, scan speed, layer thickness, and hatch distance. This approach can be explored for EB-PBF when an open-source EB-PBF system is available. Another point to note is that owing to the influence of both gravity and heat accumulation [44], the downward facing surface of an inclined strut is always rougher than an upward facing surface. The use of a different scan strategy for contours may help to mitigate heat accumulation and thus improve the finish of downward facing strut surfaces [44].

Currently, ranking the effects of voids (pores), surface roughness, and geometric errors on the mechanical properties of a lattice strut is challenging due to the difficulty of identifying the effects of each variable individually. However, the length scale of each defect may provide a useful indication. Consider a 1-mm thick strut by design. For EB-PBF of Ti-6Al-4 V, the surface roughness (R_a) is typically $\leq 40 \mu\text{m}$, while the voids are usually $\leq 80 \mu\text{m}$ when using the Arcam powder and default EB-PBF parameters. However, the resulting thinnest strut section can be $\leq 0.3 \text{ mm}$ (for small values of $\varphi_{\text{min-threshold}}$) compared to the expected 1.0 mm. It is therefore important to prevent the formation of such weakest sections, as well as to minimize surface roughness and heal the voids by HIP.

The preceding discussion has centred on uniform lattice structures. Graded lattice designs via strut size or shape variations or unit cell size or type variations has proved to be an important strategy for improving lattice stiffness, strength or functionality [7,61,62]. Ensuring robust strut additive continuity is equally crucial for these graded designs. The criterion $\varphi_{\min\text{-threshold}} = 0.90$ can easily be used to define the minimum diameter to which each strut can be tapered, with respect to strut inclination angle, layer thickness and PBF parameters. The findings of this work can thus also serve as a basic guideline for their design and manufacture.

Finally, it should be noted that the specific value of $\varphi_{\min\text{-threshold}}$ may no longer be 0.90 when Eq.(7) is applied to L-PBF due to the use of different powder feedstock materials, recoating processes, default PBF parameters and surface finishes. Separate experimental validations are required.

5. Conclusions

The following conclusions can be drawn from this study for EB-PBF of lattice materials.

- A lattice strut additive continuity model has been formulated in the form of Eq. (6) through a parameter φ , which quantifies the influence of initial layer thickness setting h_0 , strut diameter d , strut inclination angle θ and powder-bed density ρ_{PBD} on lattice strut additive continuity.
- A minimum strut additive continuity threshold value, $\varphi_{\min\text{-threshold}} \approx 0.9$, has been established, which can allow for design and manufacture of Ti-6Al-4V lattice materials with robust strut additive continuity by EB-PBF using the default Arcam EB-PBF parameters.
- The initial layer thickness setting (h_0) has a profound effect on lattice strut additive continuity. Where practical, a small h_0 is always preferred for EB-PBF of lattices with robust strut additive continuity. As a benchmark selection, a powder specific h_0 defined as $Dv(90) \times \frac{\rho_{\text{PBD}}}{\rho_{\text{TD}}}$ or approximately $0.6Dv(90)$ can be considered.
- The powder-bed layer density ρ_{PBD} can noticeably affect lattice strut additive continuity. A minimum of $\rho_{\text{PBD}} = 0.6\rho_{\text{TD}}$ is recommended for powder selection for PBF of lattice materials, where ρ_{PBD} is influenced by the powder recoating system, as well as the particle size distribution.
- The lattice strut additive continuity maps constructed in the form of Figs. 8–10 and Figs. S3–S7 provide a comprehensive guideline for the design and manufacture of lattice materials with robust strut additive continuity by EB-PBF.
- Eq. (5) analytically defines the dependence of powder-bed layer thickness on layer number (n) for PBF AM. The cut-off layer number can be set to be 10 for all powder feedstock materials with $\rho_{\text{PBD}} \geq 0.5\rho_{\text{TD}}$.

CRedit authorship contribution statement

X. Z. Zhang: Writing – original draft, Writing – review & editing, Investigation, Formal analysis, Data curation. **H. P. Tang:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition. **J. Wang:** Writing – review & editing, Data curation. **L. Jia:** Writing – review & editing, Data curation. **Y. X. Fan:** Writing – review & editing, Data curation. **M. Leary:** Writing – review & editing, Formal analysis. **Ma Qian:** Conceptualization, Writing – original draft, Writing – review & editing, Funding acquisition, Formal analysis, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.addma.2022.103022.

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